

Acoustic Signal Classification for Quality Control in Frozen Lactic Acid Cultures Using Edge AI

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Abstract— This paper presents the design and implementation of an Artificial Intelligence of Things (AIoT) based system for the non-destructive quality control of frozen lactic acid culture sachets through acoustic signal analysis. The quality of these products is highly dependent on maintaining the cold chain during storage and transport. A loss in cold chain integrity leads to structural changes in the product, resulting in a compacted texture that can be detected acoustically. The proposed system uses an embedded ESP32-S3 microcontroller with an integrated microphone to capture impact-induced acoustic signals from the sachets. A supervised machine learning model, trained on these acoustic signals, classifies the product into three categories: frozen conforming, potential cold chain breach, and non-conforming frozen product. The model is implemented on the device using Edge Impulse for real-time inference, providing immediate feedback through local actuators (LED indicators, buzzers). The system also sends results to an IoT platform for data storage, traceability, and historical reporting. Experimental validation confirms the robustness and scalability of the solution, offering an innovative, cost-effective, and automated approach for cold chain quality assurance in the food and biotechnology industries.

Keywords— AIoT, Acoustic signal analysis, Cold chain integrity, Quality control, Lactic acid cultures, Machine learning, Embedded systems

I. INTRODUCTION

Maintaining the integrity of the cold chain is crucial for the preservation of perishable products, especially in the food and biotechnology industries, where the stability and quality of the products are directly tied to their storage and transport conditions [1]. In these industries, even minor deviations from the required storage temperature can lead to irreversible damage, affecting both the safety and efficacy of the product [2].

Lactic acid cultures, which are widely used in the production of fermented dairy products, probiotics, and other biotechnological applications, are particularly sensitive to temperature fluctuations [3]. These cultures rely on a stable cold environment to maintain their viability and active properties [4].

A brief temperature excursion, even if it does not lead to immediate freezing or thawing, can significantly compromise the culture's performance, affecting both its biological and functional characteristics [5]. Even small changes in temperature can disrupt the metabolic activity of the cultures, reducing their effectiveness in fermentation processes and compromising the final product's quality and consistency [6].

Traditional cold chain monitoring systems typically rely on temperature sensors, which are reactive and only provide alerts once a threshold is exceeded. These systems fail to detect minor interruptions that may not immediately affect temperature readings but can still alter the internal structure

and quality of the product [5]. Thus, there is a need for an efficient, non-destructive, and real-time quality control method that can identify subtle changes in the product caused by cold chain disruptions.

This paper proposes an AIoT-based system using acoustic signal analysis to detect cold chain breaches in frozen lactic acid culture sachets. The solution employs an ESP32-S3 microcontroller with an integrated microphone to capture the acoustic signals generated when the sachets are impacted. These signals are then analyzed using machine learning algorithms to classify the product into three categories: conforming, potentially compromised, and non-conforming.

The system provides a cost-effective, real-time solution for monitoring product quality without the need for visual inspection or destructive testing, ensuring the cold chain integrity of the product is maintained throughout storage and transport.

Recent literature consolidates acoustic detection as a promising non-destructive alternative for inferring internal physical changes, thus supporting quality control decisions in various industries. In the agro-food sector, Aboonajmi et al. (2015) highlight the applicability of acoustic pulses or sound waves for rapid and cost-effective internal quality evaluation, correlating attributes such as crispness, firmness, and water activity with acoustic properties, emphasizing its potential for quality assessment [7].

Applied research by Audu et al. (2023) focuses on the development of two intelligent acoustic devices for classifying the quality of yams (good, damaged by disease, damaged by insects) using two excitation techniques: (i) sound generation with a speaker and (ii) controlled impact. With a limited set of variables (amplitude, frequency, intensity, etc.) and discriminant analysis as a machine learning algorithm, they report validated performance between 68–82%, demonstrating the feasibility for practical classification [8].

In additive manufacturing, a study on acoustic emission (AE) in Laser Powder Bed Fusion (L-PBF) emphasizes the potential of gas-borne acoustic monitoring for quality control, exploring machine learning to detect conditions associated with porosity. The work correlates audible emissions with keyhole parameters using a high-sampling-rate microphone [9].

In the pharmaceutical industry, a deep learning approach utilizes log-mel spectrograms (128×32) as inputs to a VGG-style CNN to classify granulation phases in a mixing process, avoiding manual feature engineering and leveraging supervised learning to capture time-frequency patterns [10].

Lastly, in process engineering, the combination of AE + ML is reported to estimate particle size distributions in gas-solid flows: regression models (e.g., Random Forest and

Gradient Boosting) are constructed using frequency and acoustic energy as inputs, demonstrating how machine learning can map noisy signals to physical properties of interest [11].

The primary contributions of this paper include the development of a novel AIoT-based system for detecting cold chain breaches in frozen lactic acid culture sachets using acoustic signal classification, this system applies machine learning to analyze acoustic signals and classify product quality in real-time. Additionally, the system is integrated with IoT platforms, specifically ThingSpeak, for traceability and remote monitoring, ensuring continuous quality assurance. Furthermore, the system's robustness and scalability are validated through experimental tests.

Overall, this work presents an innovative, cost-effective, and scalable solution for real-time cold chain monitoring, with significant potential to improve product quality and mitigate risks associated with cold chain failures in industrial and biotechnological applications.

II. SOLUTION PROPOSAL

This paper proposes the development of an AIoT-based system for detecting and monitoring cold chain breaches in frozen lactic acid culture sachets through acoustic signal classification. The system architecture consists of an ESP32-S3 microcontroller with an integrated microphone, which captures the acoustic signals generated when the sachets are subjected to a controlled impact on a rigid surface.

The hypothesis underlying the project is that a breach in the cold chain alters the internal structure of the product, which results in a distinct acoustic signature. The captured signals will undergo preprocessing, including noise filtering, normalization, and segmentation, followed by feature extraction in both the temporal and spectral domains using techniques like Mel Filterbank Energy (MFE) and Classifier.

A machine learning model will be trained to classify the sachets into three categories: conforming, potentially compromised, and non-conforming. The model will be deployed in real-time on the embedded device using the Edge Impulse platform, allowing immediate classification and feedback via local actuators such as LED indicators and buzzers.

Additionally, the system will transmit results to an IoT platform, called ThingSpeak, for traceability and remote monitoring, providing continuous quality assurance throughout the cold chain. The system's performance will be validated through experimental tests to evaluate its robustness, scalability, and accuracy in real-world conditions, ensuring its suitability for industrial and biotechnological applications. This solution offers a cost-effective, scalable, and real-time approach for maintaining cold chain integrity, improving product quality, and reducing risks associated with cold chain failures.

III. IMPLEMENTATION DETAILS

The implementation of the proposed AIoT-based system for detecting cold chain breaches in frozen lactic acid culture sachets involves several key stages, each tailored for the requirements of the system's real-time performance and scalability.

The first step was the dataset construction, acoustic data was collected using an ESP32-S3 microcontroller with an integrated microphone, capturing audio signals generated when the sachets were subjected to controlled impacts on a rigid metal surface. Each dataset was labeled with one of three categories: Descongelado (defrosted), Incierto (uncertain), and Ok (conforming), based on the product's state regarding cold chain integrity.

The dataset consisted of 157 training samples and 39 test samples, ensuring proper distribution across all classes. An 80%/20% train/test split was applied, ensuring sufficient data for training while reserving a portion for model validation.

The AI architecture was built using the Edge Impulse platform, where the extracted features were fed into a classifier block. A supervised machine learning model was trained to classify the three product states: Descongelado (defrosted), Incierto (uncertain), and Ok (conforming).

Mel-Filterbank Energy (MFE) was selected as the preprocessing method to extract features from the acoustic signals generated when the freeze-dried cultures were subjected to a controlled impact. MFE computes the spectral energy distribution of an audio signal across a set of filters spaced according to the Mel frequency scale, which approximates the nonlinear frequency sensitivity of the human auditory system. The main reason for choosing MFE is that it preserves the energy distribution across frequency bands without applying additional transformations such as the discrete cosine transform used in Mel Frequency Cepstral Coefficients (MFCC). This property makes MFE particularly suitable for tasks involving non-speech acoustic events, where the relevant information is primarily contained in the spectral energy patterns rather than in cepstral coefficients.

In this project, the classification task relies on identifying differences in the acoustic response produced by freeze-dried cultures depending on their structural condition. These differences manifest as variations in frequency-dependent energy patterns generated by the impact sound. MFE features allow these patterns to be captured directly and represented in a compact form suitable for neural network training. Additionally, MFE offers several practical advantages for edge-AI applications. The representation is computationally efficient, produces a relatively low-dimensional feature vector, and is compatible with lightweight convolutional neural network architectures commonly used in embedded machine learning frameworks such as Edge Impulse. This efficiency enables real-time inference on resource-constrained devices such as the XIAO ESP32S3 Sense.

The classification stage was implemented using the classification learning block provided by the Edge Impulse platform, which generates a lightweight neural network architecture optimized for embedded machine learning applications. This block automatically constructs and trains a neural network designed to learn discriminative patterns in the input feature space and map them to discrete class labels.

The architecture generated by the classification block consists of a convolutional neural network (CNN) designed for time-series data. Convolutional layers are used to detect local patterns in the feature representation, such as frequency-dependent energy variations, while pooling and dropout layers help improve generalization and reduce the risk of overfitting. The final output layer uses a softmax activation function to

produce probability scores for each class, allowing the model to determine the most likely state of the culture sample.

Model training was performed for 100 training cycles with a learning rate of 0.005, producing a compact model suitable for deployment on the XIAO ESP32S3 Sense microcontroller. After training, the model was quantized to 8-bit integer precision (int8), reducing memory usage and enabling efficient real-time inference on the embedded device. This approach allows the system to perform on-device classification of acoustic events, eliminating the need for continuous cloud processing and enabling rapid decision-making directly on the edge device.

The neural network architecture used in this study was automatically generated and optimized through the Edge Impulse classification pipeline for deployment on resource-constrained embedded hardware. The input layer receives a feature vector of 4018 elements, representing the spectral energy distribution extracted from the audio signal. These features are reshaped into a two-dimensional representation to enable convolutional operations across the temporal-spectral domain. The first convolutional layer applies eight convolutional filters with a kernel size of three, allowing the network to detect localized patterns in the spectral energy distribution. These filters learn to identify fundamental acoustic structures present in the impact sounds generated by the cultures.

A dropout layer with a rate of 0.25 follows this stage to reduce the likelihood of overfitting by randomly disabling a fraction of neurons during training. This encourages the network to learn more robust and generalized feature representations. The second convolutional layer increases the number of filters to sixteen, enabling the model to capture more complex acoustic patterns that may be associated with the structural condition of the freeze-dried samples. This hierarchical structure allows the network to progressively learn higher-level representations of the acoustic signal.

After convolutional processing, a flatten layer converts the extracted feature maps into a one-dimensional vector that can be used by the final classification stage. The output layer then performs the final decision process, producing probability scores for each class through a softmax activation function.

Following the development and training of the classification model within the Edge Impulse platform, the trained model was deployed to an embedded hardware environment to enable real-time inference. As previously mentioned, the implementation was carried out on a XIAO ESP32-S3 Sense microcontroller, which integrates a PDM microphone and provides sufficient computational resources for executing lightweight machine learning models directly at the edge.

The firmware continuously acquires audio samples from the onboard microphone using the I2S interface. Audio data are collected in fixed-length buffers corresponding to the input window size required by the deployed model. Prior to executing inference, the system performs a preliminary acoustic event detection step based on the average signal energy of the captured samples. This mechanism functions as a trigger to ensure that the classifier is activated only when a relevant acoustic event—such as the impact applied to the culture container—is detected. This strategy reduces unnecessary processing and improves computational efficiency.

When the measured signal energy exceeds the predefined threshold, the buffered audio window is processed by the embedded Edge Impulse inference engine. The classifier generates probability scores for each of the three predefined classes: *Ok*, *Descongelado*, and *Incierto*. The algorithm then identifies the class with the highest probability and computes the difference between this value and that of the second most probable class. This confidence margin, together with predefined probability thresholds, is used to enhance the robustness of the decision-making process and reduce the likelihood of uncertain classifications.

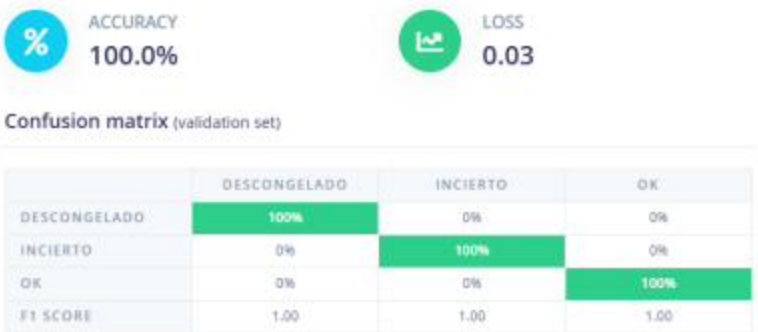
Based on the final classification result, the system activates different local responses through onboard actuators. A short LED blink indicates the *Ok* state, while a continuous high-frequency acoustic alarm signals the *Descongelado* condition, which may indicate a potential disruption in the cold chain. If the prediction does not meet the required confidence criteria, an intermittent low-frequency alarm is triggered to represent the *Incierto* state.

In addition to local feedback, the system can transmit classification results and associated metrics, such as prediction confidence and acoustic event intensity, to a cloud-based IoT platform, ThingSpeak, for remote monitoring and data logging using the local WiFi. This integration enables real-time tracking of potential cold-chain integrity issues while maintaining inference at the edge, thereby minimizing latency and reducing dependence on continuous network connectivity.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The model was trained using the Edge Impulse platform, achieving 100% accuracy on the validation set, the confusion matrix showed perfect classification for all three categories: *Descongelado* (defrosted), *Incierto* (uncertain), and *Ok* (conforming). The model demonstrated high performance in real-time with an inference time of just 6 ms per sample, allowing it to operate efficiently on the embedded device. The loss was reported at 0.03, reflecting minimal deviation between predicted and actual classifications.

Image 1. Confusion Matrix



The system's robustness was confirmed through various performance metrics, including the area under the ROC curve and the weighted averages for precision, recall, and F1 score, all of which were recorded as 1.00, indicating optimal classification performance.

The model was deployed on the ESP32-S3 microcontroller, where it maintained a flash usage of 40.9K and RAM usage of 14.9K, ensuring that the solution was lightweight and suitable for embedded systems.

Additionally, the system provided real-time feedback through LED indicators and buzzers, based on the

classification results. The results were also transmitted to an IoT platform for remote monitoring and traceability, ensuring continuous quality assurance throughout the cold chain.

Although the confusion matrix obtained during model evaluation in the Edge Impulse environment indicated a classification accuracy of 100%, the performance observed during deployment on the XIAO ESP32-S3 Sense revealed that the system still presented considerable room for improvement under real operating conditions. In practice, the real-time classification accuracy was lower than expected, suggesting that the model's apparent performance during validation did not fully translate to the embedded environment. A key contributing factor was the lack of a clearly defined acoustic noise threshold for event triggering, which affected the reliability of the initial sound detection stage. Additionally, the system did not incorporate a well-characterized baseline for ambient silence, meaning that the calibration of the background noise level was not sufficiently established prior to inference. As a result, variations in environmental noise and impact intensity sometimes led to inconsistent triggering of the classifier or ambiguous predictions. These observations highlight the importance of carefully calibrating the event detection parameters and background noise profile when deploying acoustic machine learning models in real-world embedded systems, as evaluation metrics obtained in controlled datasets may not fully reflect operational performance.

V. CONCLUSIONS

This paper presents an AIoT-based system designed for the non-destructive quality control of frozen lactic acid culture sachets using acoustic signal classification. The system demonstrated outstanding performance, achieving 100% accuracy in classifying product states into conforming, potentially compromised, and non-conforming categories based on the internal structure of the sachets on the training phase.

The use of machine learning algorithms for analyzing acoustic signals proved to be highly effective in detecting subtle changes in the product, which are indicative of cold chain breaches. By incorporating acoustic analysis with real-time machine learning inference, the system provides a robust, scalable, and cost-effective solution for continuous monitoring and quality assurance.

The model, deployed on the ESP32-S3 microcontroller, performed exceptionally well with an inference time of just 6 ms per sample, which makes the system suitable for embedded applications in industrial environments where real-time decision-making is crucial. The compact nature of the system, combined with its low RAM usage (14.9K) and efficient flash usage (40.9K), ensures its feasibility for integration into various IoT-enabled platforms for remote monitoring and traceability.

The integration with an IoT platform enables the system to send results for continuous quality assurance, allowing for the real-time monitoring of cold chain conditions and immediate feedback to users through local actuators such as LED indicators and buzzers.

This work demonstrates the feasibility of implementing an edge-based acoustic classification system for assessing the condition of frozen lactic cultures using a lightweight machine learning model deployed on a XIAO ESP32-S3 Sense

microcontroller. The integration of embedded inference with local actuation and IoT connectivity illustrates the potential of AIoT solutions for real-time monitoring in cold-chain applications. Although the model achieved perfect classification performance during evaluation within the development environment, the results obtained during on-device deployment revealed that real-world performance still requires further optimization. In particular, the absence of a well-defined acoustic noise threshold and the lack of a calibrated baseline for ambient silence affected the reliability of event detection and, consequently, the consistency of the classification results. These findings highlight the importance of robust signal preprocessing and environmental calibration when transferring machine learning models from controlled training conditions to embedded real-world implementations. Future work should focus on refining the acoustic triggering mechanism, improving noise calibration, and expanding the dataset under realistic conditions to enhance the robustness and reliability of the system.

This research contributes to the growing body of knowledge regarding the application of AIoT and acoustic monitoring systems for quality control. It paves the way for future advancements in cold chain management and product integrity monitoring.

Further work will focus on improving the system's ability to handle more complex and dynamic environments, incorporating additional sensors such as temperature and humidity for multivariate quality monitoring. Additionally, efforts will be directed toward expanding the system's applicability to other types of perishable products in various industrial settings.

The proposed solution offers a scalable, non-destructive, and efficient method for ensuring cold chain integrity, ultimately enhancing product quality, reducing waste, and increasing consumer trust in the food and biotechnology sectors.

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